



PORTFOLIO EDITION

BLUE TEAM INTERNSHIP

Week 8 Task Report

Windows Artifact Analysis

Event Logs · Prefetch · Thumbcache · Recycle Bin

Prepared by Malaika Azhar · Week 8 (Days 53–59)

Table of Contents

01	Executive Summary	3
02	Week 8 Objectives	4
03	Days 53-55 — Event Log Analysis	5
04	Days 56-57 — Prefetch File Analysis	10
05	Days 58-59 — Cache, Thumbcache & Recycle Bin	14
06	Technical Forensic Summary	18
07	Executive Summary (Leadership)	19
08	Key Learnings & Reflection	20
09	Closing Statement & Sign-Off	21
A	<i>Appendix — Deliverables & Screenshot Index</i>	22

Executive Summary

Overview of Week 8 Deliverables

This report documents Week 8 of the Blue Team Internship, covering Days 53 through 59. Where earlier weeks in Phase Two established the principles of digital forensics — chain of custody, imaging, and hashing — Week 8 moved into hands-on live endpoint artefact extraction across three categories of Windows forensic evidence: Security event logs, Prefetch execution records, and cache/thumbnaill artefacts including the Recycle Bin.

All work was performed against the intern's own live, personal Windows machine, using Eric Zimmerman's EZ Tools suite (EvtxECmd and PECmd) for structured parsing, Thumbcache Viewer for thumbnail extraction, and BrowsingHistoryView for combined Chrome and Edge browser cache analysis. Every extraction was preceded by a SHA256 integrity hash of the source artefact, and every finding was cross-referenced against the intern's own known activity to separate genuine human behaviour from expected Windows background noise.

The Security event log baseline confirmed the intern's physical logon pattern via Logon Type 2 events, captured one failed-logon (Event ID 4625) instance, and — following a deliberate screen lock/unlock test — still returned no Logon Type 7 (unlock) events in the parsed output, which is documented explicitly as a finding rather than omitted. Privilege elevation (Event ID 4672) was traced predominantly to the NT AUTHORITY\SYSTEM account during system startup, and background noise was confirmed via Logon Type 5 service accounts. Prefetch analysis of 363 copied .pf files identified the top 20 most-executed programs (led by Chrome and core Windows services), confirmed no suspicious execution from Temp or Downloads among the top-run entries, and separately identified an installer temp-path execution (Advanced SystemCare Setup) that was investigated and explained as expected InstallShield behaviour rather than flagged as malicious. Thumbcache, browser cache, and Recycle Bin extraction closed out the week, recovering 13 thumbnail entries, a combined 568-record Chrome/Edge history export, and 74 paired \$I/\$R Recycle Bin deletion records — with all personal filenames and URLs withheld from this report per the privacy handling standard set out in the task brief.

Task Block	Status	Notes
Days 53-55 — Event Log Analysis	Complete	Human baseline, failed logon, privilege events, and service-account noise all documented
Days 56-57 — Prefetch Analysis	Complete	363 .pf files parsed, top-20 programs profiled, one explained Temp-path installer execution
Days 58-59 — Cache & Recycle Bin	Complete	Thumbcache, browser cache, and Recycle Bin all extracted and documented

As with prior internship reports, this document records the practical friction encountered along the way — including tool download and extraction issues, locked thumbcache database files requiring an Explorer restart workaround, and a Logon Type 7 event that did not appear even after a deliberate lock/unlock test — rather than presenting the week as friction-free.

Week 8 Objectives

Days 53–59

Week 7 introduced the principles of digital forensics: chain of custody, imaging, hashing, and the legal and procedural foundations that make evidence defensible. Week 8 moved from principles into hands-on artefact extraction and interpretation across three categories of Windows forensic evidence: event logs, prefetch files, and cache and thumbnail data — all extracted live from the intern's own personal machine rather than from a pre-built evidence file.

Learning Outcomes Targeted

- Identify and explain the significance of the key Windows Security event IDs related to authentication and account management (4624, 4625, 4672, 1102), and distinguish Logon Types 2, 5, 7, and 10.
- Parse Windows Prefetch files to build an execution timeline, and assess whether an execution path (System32, Program Files, Temp, Downloads, AppData) is expected or warrants further investigation.
- Extract thumbnail and browser cache artefacts from a live system and explain why they persist independently of the files or browsing sessions that generated them.
- Understand what the \$I and \$R files in the Recycle Bin contain individually, and how together they constitute a complete deletion record (original path, deletion timestamp, and recoverable data).
- Practise the core forensic skill of correlation: connecting findings across event logs, prefetch, and cache artefacts into a single coherent narrative rather than three isolated reports.

SCOPE NOTE

This report follows the three-block structure specified in the Week 8 task brief: Days 53–55 (Event Log Analysis), Days 56–57 (Prefetch File Analysis), and Days 58–59 (Cache, Thumbcache, and Recycle Bin Analysis). Each section documents what was done, what was observed, what was concluded, and any troubleshooting undertaken along the way.

Days 53-55 — Event Log Analysis

Concepts: chain of custody · Logon Types 2/5/7 · privilege escalation (4672) · background service noise

Objective

Extract and analyse the intern's own live Windows Security event log (EVTX format) to establish an authentication baseline: physical logon/unlock activity (Day 53), privilege elevation events (Day 54), and background service-account noise (Day 55) — using Eric Zimmerman's EvtxECmd to parse the raw log into a structured, filterable CSV.

Evidence Integrity — Source Log Copy & Hash

The live Security.evtx log was copied from C:\Windows\System32\winevt\Logs\ into a dedicated working folder (C:\Forensics\EventLogs\) under an elevated (Administrator) terminal, and a SHA256 hash was generated immediately after the copy to serve as the evidence integrity record for this task.

```
mkdir C:\Forensics\EventLogs
copy C:\Windows\System32\winevt\Logs\Security.evtx C:\Forensics\EventLogs\Security.evtx
certutil -hashfile C:\Forensics\EventLogs\Security.evtx SHA256
```

```
Administrator: Windows PowerShell
CertUtil: -hashfile command FAILED: 0x80070002 (WIN32: 2 ERROR_FILE_NOT_FOUND)
CertUtil: The system cannot find the file specified.
PS C:\Windows\system32> mkdir C:\Forensics\EventLogs

Directory: C:\Forensics

Mode                LastWriteTime         Length Name
----                -
d-----          9/12/2026   2:34 AM                EventLogs

PS C:\Windows\system32> copy C:\Windows\System32\winevt\Logs\Security.evtx C:\Forensics\EventLogs\Security.evtx
PS C:\Windows\system32> certutil -hashfile C:\Forensics\EventLogs\Security.evtx SHA256
SHA256 hash of C:\Forensics\EventLogs\Security.evtx:
b84b2dc49f85b66ff99b9173d80dc77cee63f39455b552250c77610de36b9463
CertUtil: -hashfile command completed successfully.
PS C:\Windows\system32>
```

Figure 53.1 — certutil SHA256 output confirming the integrity hash of the copied Security.evtx file: b84b2dc4...36b9463.

Parsing with EvtxECmd

EvtxECmd (Eric Zimmerman's EZ Tools) was downloaded and run against the copied log to produce a structured CSV for filtering in a spreadsheet. Two separate parsing passes were carried out over the course of the investigation, the second following a deliberate screen lock/unlock test (see Day 53 below).

```
.\EvtxECmd.exe -f C:\Forensics\EventLogs\Security.evtx --csv C:\Forensics\EventLogs\ --csvf output.csv
```

```
PS C:\Windows\system32> cd "C:\Users\hp\Downloads\EvtxECmd\EvtxECmd"
PS C:\Users\hp\Downloads\EvtxECmd\EvtxECmd> .\EvtxECmd.exe -f C:\Forensics\EventLogs\Security.evtx --csv C:\Forensics\Ev
entLogs\ --csvf output.csv
EvtxECmd version 2026.5.0

Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/evttx

Command line: -f C:\Forensics\EventLogs\Security.evtx --csv C:\Forensics\EventLogs\ --csvf output.csv

CSV output will be saved to C:\Forensics\EventLogs\output.csv
```

Figure 53.2 — EvtxECmd confirming CSV output will be saved to C:\Forensics\EventLogs\output.csv.

Figure 53.2b — *EvtxECmd* Pass 1 completion log — 32,738 records included, 0 errors, 0 dropped events.

Parsing Pass	Total Records	Event 4624	Event 4625	Event 4672
Pass 1 (output.csv)	32,738	1,237	242	1,160
Pass 2 (output2.csv, post lock/unlock)	33,127	1,049	290	986

Day 53 — The Human Baseline

Filtering Event ID 4624 for Logon Type 2 (interactive logon at the physical console) and Logon Type 7 (screen unlock) was used to establish the intern's physical working-hours pattern, as required by the task brief.

File

Home

Insert

Page Layout

Formulas

Data

Review

View

New

Open

Save

Save As

Quick Print

Print

Print Preview

Undo

Redo

Convert PDF to XLSX

DOCX Editor

DOCX and XLSX Reader

Encrypt with Password

Document Properties

Show File Info

Common

Toolset

Info

Explorer

output.csv

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	RecordId	EventRecId	TimeCreate	EventId	Level	Provider	Channel	ProcessId	ThreadId	Computer	ChunkNum	UserSid	MapDescr	UserSnm	RemoteH	PayloadD	PayloadD	PayloadD	PayloadD	Pay
2	231733	231733	#####		4672	LogAlway	Microsoft: Security	828	9024	DESKTOP-	0		Administ	NT AUTHORITY\SYST	Privilege: Logonid: 0x3E7					
3	231734	231734	#####		4624	LogAlway	Microsoft: Security	828	9024	DESKTOP-	0		Successfu	WORKGR- (-)	Target: NT LogonTyp: Logonid: 0x3E7					
4	231735	231735	#####		4672	LogAlway	Microsoft: Security	828	9024	DESKTOP-	0		Administ	NT AUTHORITY\SYST	Privilege: Logonid: 0x3E7					
5	231736	231736	#####		4624	LogAlway	Microsoft: Security	828	9024	DESKTOP-	0		Successfu	WORKGR- (-)	Target: NT LogonTyp: Logonid: 0x3E7					
6	231737	231737	#####		4672	LogAlway	Microsoft: Security	828	9024	DESKTOP-	0		Administ	NT AUTHORITY\SYST	Privilege: Logonid: 0x3E7					
7	231738	231738	#####		4799	LogAlway	Microsoft: Security	828	9024	DESKTOP-	0		A security	WORKGROUP\DESKT	Target: Bu Subjecto: CallerProc: CallerProcessId: 0x970					
8	231739	231739	#####		4624	LogAlway	Microsoft: Security	828	4300	DESKTOP-	0		Successfu	WORKGR- (-)	Target: NT LogonTyp: Logonid: 0x3E7					
9	231740	231740	#####		4672	LogAlway	Microsoft: Security	828	4300	DESKTOP-	0		Administ	NT AUTHORITY\SYST	Privilege: Logonid: 0x3E7					
10	231741	231741	#####		5379	LogAlway	Microsoft: Security	828	4300	DESKTOP-	0		Credentia	DESKTOP-003U1SH\$	SID: S-1-5-Domain: V Logonid: (CountOfCredentia	Ac				
11	231742	231742	#####		5379	LogAlway	Microsoft: Security	828	4300	DESKTOP-	0		Credentia	DESKTOP-003U1SH\$	SID: S-1-5-Domain: V Logonid: (CountOfCredentia	Ac				
12	231743	231743	#####		5379	LogAlway	Microsoft: Security	828	4300	DESKTOP-	0		Credentia	DESKTOP-003U1SH\$	SID: S-1-5-Domain: V Logonid: (CountOfCredentia	Ac				
13	231744	231744	#####		5379	LogAlway	Microsoft: Security	828	4300	DESKTOP-	0		Credentia	DESKTOP-003U1SH\$	SID: S-1-5-Domain: V Logonid: (CountOfCredentia	Ac				
14	231745	231745	#####		5379	LogAlway	Microsoft: Security	828	4300	DESKTOP-	0		Credentia	DESKTOP-003U1SH\$	SID: S-1-5-Domain: V Logonid: (CountOfCredentia	Ac				
15	231746	231746	#####		5379	LogAlway	Microsoft: Security	828	1468	DESKTOP-	0		Credentia	DESKTOP-003U1SH\$	SID: S-1-5-Domain: V Logonid: (CountOfCredentia	Ac				
16	231747	231747	#####		5379	LogAlway	Microsoft: Security	828	1468	DESKTOP-	0		Credentia	DESKTOP-003U1SH\$	SID: S-1-5-Domain: V Logonid: (CountOfCredentia	Ac				
17	231748	231748	#####		5379	LogAlway	Microsoft: Security	828	1468	DESKTOP-	0		Credentia	hp	SID: S-1-5-Domain: L Logonid: (CountOfCredentia	Ac				
18	231749	231749	#####		5379	LogAlway	Microsoft: Security	828	1468	DESKTOP-	0		Credentia	hp	SID: S-1-5-Domain: L Logonid: (CountOfCredentia	Ac				
19	231750	231750	#####		5379	LogAlway	Microsoft: Security	828	1468	DESKTOP-	0		Credentia	hp	SID: S-1-5-Domain: L Logonid: (CountOfCredentia	Ac				
20	231751	231751	#####		5379	LogAlway	Microsoft: Security	828	1468	DESKTOP-	0		Credentia	hp	SID: S-1-5-Domain: L Logonid: (CountOfCredentia	Ac				
21	231752	231752	#####		4624	LogAlway	Microsoft: Security	828	4300	DESKTOP-	0		Successfu	WORKGR- (-)	Target: NT LogonTyp: Logonid: 0x3E7					
22	231753	231753	#####		4672	LogAlway	Microsoft: Security	828	4300	DESKTOP-	0		Administ	NT AUTHORITY\SYST	Privilege: Logonid: 0x3E7					
23	231754	231754	#####		4624	LogAlway	Microsoft: Security	828	4300	DESKTOP-	0		Successfu	WORKGR- (-)	Target: NT LogonTyp: Logonid: 0x3E7					

Figure 53.3 — Parsed output.csv opened in a spreadsheet, filtered on MapDescription and Logon Type columns ahead of the Logon Type 2 search.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	
1446	233177	233177	7/24/2026 23:25	4798	LogAlwa	MicrosoftSecurity		828	3396	DESKTOP	13		A user's h	WORKGROUP\DES	Target: D	SubjectLogonId: 0x3E7	CallerProc	CallerProcessId: 0x1D70				FALSE	C:\Foren\Audit suc	0	0	0	
1447	233178	233178	7/24/2026 23:25	1100	Info	MicrosoftSecurity		1240	6932	DESKTOP	13		The event logging service has shut do									FALSE	C:\Foren\Audit suc	0	0	0	
1448	233179	233179	7/24/2026 23:26	4688	LogAlwa	MicrosoftSecurity		4	32	DESKTOP	13		A new pri	-	Parent p	PID: 0x4	Parent PI	Mandato	Target User: -	Registry	(PID: 0x4	FALSE	C:\Foren\Audit suc	0	0	0	
1449	233180	233180	7/24/2026 23:26	4696	LogAlwa	MicrosoftSecurity		4	32	DESKTOP	13		Apri	privileg	-	Target: -	(S-1-5-18)					FALSE	C:\Foren\Audit suc	1168	0	0	
1450	233181	233181	7/24/2026 23:26	4826	LogAlwa	MicrosoftSecurity		4	32	DESKTOP	13											FALSE	C:\Foren\Audit suc	0	0	0	
1451	233182	233182	7/24/2026 23:26	4688	LogAlwa	MicrosoftSecurity		4	32	DESKTOP	13		A new pri	-	Parent p	PID: 0x4	Parent PI	Mandato	Target User: -	C:\Windo	FALSE	C:\Foren\Audit suc	0	0	0		
1452	233183	233183	7/24/2026 23:26	4688	LogAlwa	MicrosoftSecurity		4	332	DESKTOP	13		A new pri	-	Parent p	PID: 0x184	Parent PI	Mandato	Target User: -	C:\Windo	FALSE	C:\Foren\Audit suc	0	0	0		
1453	233184	233184	7/24/2026 23:26	4688	LogAlwa	MicrosoftSecurity		4	196	DESKTOP	13		A new pri	-	Parent p	PID: 0x184	Parent PI	Mandato	Target User: -	C:\Windo	FALSE	C:\Foren\Audit suc	0	0	0		
1454	233185	233185	7/24/2026 23:27	4688	LogAlwa	MicrosoftSecurity		4	393	DESKTOP	13		A new pri	-	Parent p	PID: 0x200	Parent PI	Mandato	Target User: -	C:\Windo	FALSE	C:\Foren\Audit suc	0	0	0		
1455	233186	233186	7/24/2026 23:27	4688	LogAlwa	MicrosoftSecurity		4	388	DESKTOP	13		A new pri	-	Parent p	PID: 0x184	Parent PI	Mandato	Target User: -	C:\Windo	FALSE	C:\Foren\Audit suc	0	0	0		
1456	233187	233187	7/24/2026 23:27	4688	LogAlwa	MicrosoftSecurity		4	388	DESKTOP	13		A new pri	-	Parent p	PID: 0x298	Parent PI	Mandato	Target User: -	C:\Windo	FALSE	C:\Foren\Audit suc	0	0	0		
1457	233188	233188	7/24/2026 23:27	4688	LogAlwa	MicrosoftSecurity		4	388	DESKTOP	13		A new pri	-	Parent p	PID: 0x200	Parent PI	Mandato	Target User: -	C:\Windo	FALSE	C:\Foren\Audit suc	0	0	0		
1458	233189	233189	7/24/2026 23:27	4688	LogAlwa	MicrosoftSecurity		4	116	DESKTOP	13		A new pri	-	Parent p	PID: 0x2AC	Parent PI	Mandato	Target User: -	C:\Windo	FALSE	C:\Foren\Audit suc	0	0	0		
1459	233190	233190	7/24/2026 23:27	4688	LogAlwa	MicrosoftSecurity		4	116	DESKTOP	13		A new pri	-	Parent p	PID: 0x298	Parent PI	Mandato	Target User: -	C:\Windo	FALSE	C:\Foren\Audit suc	0	0	0		
1460	233191	233191	7/24/2026 23:27	4688	LogAlwa	MicrosoftSecurity		4	116	DESKTOP	13		A new pri	-	Parent p	PID: 0x2AC	Parent PI	Mandato	Target User: -	C:\Windo	FALSE	C:\Foren\Audit suc	0	0	0		
1461	233192	233192	7/24/2026 23:27	4608	LogAlwa	MicrosoftSecurity		792	796	DESKTOP	13		Windows is starting up									FALSE	C:\Foren\Audit suc	0	0	0	
1462	233193	233193	7/24/2026 23:27	4624	LogAlwa	MicrosoftSecurity		792	796	DESKTOP	13		Successf	-	-	Target: N	LogonType 0	LogonId: 0x3E7				FALSE	C:\Foren\Audit suc	0	0	0	
1463	233194	233194	7/24/2026 23:27	4902	LogAlwa	MicrosoftSecurity		792	892	DESKTOP	13											FALSE	C:\Foren\Audit suc	0	0	0	
1464	233195	233195	7/24/2026 23:27	4648	LogAlwa	MicrosoftSecurity		792	924	DESKTOP	13		A logon w	WORKGR	-	Target: R	TargetServerName: l	PID: 0x2AC				C:\Windo	FALSE	C:\Foren\Audit suc	0	0	0
1465	233196	233196	7/24/2026 23:27	4624	LogAlwa	MicrosoftSecurity		792	924	DESKTOP	13		Successf	WORKGR	-	Target: R	LogonType 2	LogonId: 0x158DC				C:\Windo	FALSE	C:\Foren\Audit suc	0	0	0
1466	233197	233197	7/24/2026 23:27	4624	LogAlwa	MicrosoftSecurity		792	924	DESKTOP	13		Successf	WORKGR	-	Target: N	LogonType 5	LogonId: 0x3E7				C:\Windo	FALSE	C:\Foren\Audit suc	0	0	0
1467	233198	233198	7/24/2026 23:27	4672	LogAlwa	MicrosoftSecurity		792	924	DESKTOP	13		Administ	NT AUTHORITY\SYST	Privilege	LogonId: 0x3E7						FALSE	C:\Foren\Audit suc	0	0	0	
1468	233199	233199	7/24/2026 23:27	4648	LogAlwa	MicrosoftSecurity		792	928	DESKTOP	13		A logon w	WORKGR	-	Target: R	TargetServerName: l	PID: 0x310				C:\Windo	FALSE	C:\Foren\Audit suc	0	0	0
1469	233200	233200	7/24/2026 23:27	4624	LogAlwa	MicrosoftSecurity		792	928	DESKTOP	13		Successf	WORKGR	-	Target: R	LogonType 2	LogonId: 0x158E9				C:\Windo	FALSE	C:\Foren\Audit suc	0	0	0
1470	233201	233201	7/24/2026 23:27	4624	LogAlwa	MicrosoftSecurity		792	928	DESKTOP	14		Successf	WORKGR	-	Target: N	LogonType 5	LogonId: 0x3E4				C:\Windo	FALSE	C:\Foren\Audit suc	0	0	0
1471	233202	233202	7/24/2026 23:27	4672	LogAlwa	MicrosoftSecurity		792	928	DESKTOP	14		Administ	NT AUTHORITY\NET	Privilege	LogonId: 0x3E4						FALSE	C:\Foren\Audit suc	0	0	0	
1472	233203	233203	7/24/2026 23:27	4624	LogAlwa	MicrosoftSecurity		792	888	DESKTOP	14		Successf	WORKGR	-	Target: N	LogonType 5	LogonId: 0x3E7				C:\Windo	FALSE	C:\Foren\Audit suc	0	0	0
1473	233204	233204	7/24/2026 23:27	4672	LogAlwa	MicrosoftSecurity		792	888	DESKTOP	14		Administ	NT AUTHORITY\SYST	Privilege	LogonId: 0x3E7						FALSE	C:\Foren\Audit suc	0	0	0	
1474	233205	233205	7/24/2026 23:27	4624	LogAlwa	MicrosoftSecurity		792	928	DESKTOP	14		Successf	WORKGR	-	Target: N	LogonType 5	LogonId: 0x3E5				C:\Windo	FALSE	C:\Foren\Audit suc	0	0	0
1475	233206	233206	7/24/2026 23:27	4672	LogAlwa	MicrosoftSecurity		792	928	DESKTOP	14		Administ	NT AUTHORITY\LOC	Privilege	LogonId: 0x3E5						FALSE	C:\Foren\Audit suc	0	0	0	
1476	233207	233207	7/24/2026 23:27	4624	LogAlwa	MicrosoftSecurity		792	928	DESKTOP	14		Successf	WORKGR	-	Target: N	LogonType 5	LogonId: 0x3E7				C:\Windo	FALSE	C:\Foren\Audit suc	0	0	0
1477	233208	233208	7/24/2026 23:27	4672	LogAlwa	MicrosoftSecurity		792	928	DESKTOP	14		Administ	NT AUTHORITY\SYST	Privilege	LogonId: 0x3E7						FALSE	C:\Foren\Audit suc	0	0	0	
1478	233209	233209	7/24/2026 23:27	4624	LogAlwa	MicrosoftSecurity		792	888	DESKTOP	14		Successf	WORKGR	-	Target: N	LogonType 5	LogonId: 0x3E7				C:\Windo	FALSE	C:\Foren\Audit suc	0	0	0
1479	233210	233210	7/24/2026 23:27	4672	LogAlwa	MicrosoftSecurity		792	888	DESKTOP	14		Administ	NT AUTHORITY\SYST	Privilege	LogonId: 0x3E7						FALSE	C:\Foren\Audit suc	0	0	0	
H	A	B	output																								

Figure 53.4 — First identified LogonType 2 (interactive console login) event — EventId 4624, TimeCreated 7/24/2026 23:27.

Finding — Logon Type 7 (Screen Unlock)

No Logon Type 7 events were present in the initial log export. Per the task brief's contingency instruction, the screen was deliberately locked and re-unlocked with the correct password to generate a fresh unlock event, the live Security log was re-copied, and EvtxECmd was re-run to produce a second export (output2.csv, 33,127 records). Despite this deliberate test, no 'LogonType 7' string was found on repeated search of the reparsed log. This absence is documented here as a genuine finding rather than omitted or fabricated: it suggests either that this authentication method on this machine does not register as Type 7 in the standard field position, or that the specific unlock mechanism used (PIN/Windows Hello vs. password) is logged differently. This is noted as an item for further investigation rather than asserted as a resolved result.

```
Administrator: Command Prompt
Record #: 231129 (timestamp: 2026-07-24 21:34:37.2865858): Warning! Time just went backwards! Last seen time before change: 2026-09-11 23:23:55.3830215
Event log details
Flags: IsDirty
Chunk count: 320
Stored/Calculated CRC: B85C63E9/B85C63E9
Earliest timestamp: 2026-07-24 21:34:37.2865858
Latest timestamp: 2026-09-13 08:28:38.6789862
Total event log records found: 33,127
Records included: 33,127 Errors: 0 Events dropped: 0
Metrics (including dropped events)
Event ID Count
1188 9
1191 4
4688 12
4616 5
4624 1,049
4625 296
4634 16
4647 9
4648 46
4672 986
4688 123
4696 12
4707 15
4708 15,470
4799 583
4826 12
4862 12
4884 3
4885 3
5824 12
5833 12
5838 1
5858 62
5859 37
5861 62
5379 11,329
5382 18
Processed 1 file in 437.0692 seconds
```

Figure 53.4b — EvtxECmd Pass 2 completion log, run against the reparsed Security.vtx following the deliberate lock/unlock test — 33,127 records included, latest timestamp 2026-09-13 08:28:38.

Failed Logon — Event ID 4625

Event ID 4625 (failed logon) was located in the reparsed log (output2.csv). The most recent failed-logon instance was identified with its timestamp and associated failure-reason field.

[illegible]

Day 54 — Privilege Tracking and UAC Auditing

Figure 54.1 — screenshot not yet supplied (Day54_Step1_PrivilegeEvents.png).

Interpretation — SYSTEM-Tied 4672 Events

Day 55 — Demystifying the Background Noise

M	N	O	P	Q	R	S	T	U	V	W	X	Y
MapDescription	UserName	RemoteHost	PayloadData1	PayloadData2	PayloadData3	PayloadData4	PayloadData5	PayloadData6	ExecutablePath	HiddenResources	SourceFile	Keywords
Administrative logon	NT AUTHORITY\SYSTEM (S-1-5-18)		PrivilegeL	LogonId: 0x3E7						FALSE	C:\Forens	Audit succ
Successful logon	WORKGROUP\DESKTOP-0031- (-)		Target: NT	LogonType 5	LogonId: 0x3E7				C:\Window	FALSE	C:\Forens	Audit succ
Administrative logon	NT AUTHORITY\SYSTEM (S-1-5-18)		PrivilegeL	LogonId: 0x3E7						FALSE	C:\Forens	Audit succ
Successful logon	WORKGROUP\DESKTOP-0031- (-)		Target: NT	LogonType 5	LogonId: 0x3E7				C:\Window	FALSE	C:\Forens	Audit succ
Administrative logon	NT AUTHORITY\SYSTEM (S-1-5-18)		PrivilegeL	LogonId: 0x3E7						FALSE	C:\Forens	Audit succ
A security-enabled l	WORKGROUP\DESKTOP-003U1SH\$ (S-1-5-18)		Target: Bu	SubjectLogonId: 0x3E7	CallerProc	CallerProcessId: 0x970				FALSE	C:\Forens	Audit succ
Successful logon	WORKGROUP\DESKTOP-0031- (-)		Target: NT	LogonType 5	LogonId: 0x3E7				C:\Window	FALSE	C:\Forens	Audit succ
Administrative logon	NT AUTHORITY\SYSTEM (S-1-5-18)		PrivilegeL	LogonId: 0x3E7						FALSE	C:\Forens	Audit succ
Credential Manager (	DESKTOP-003U1SH\$		SID: S-1-5- Domain: WORKGROU	LogonId: (CountOfCredentials ActivityID	Microsoft					FALSE	C:\Forens	Audit succ
Credential Manager (	DESKTOP-003U1SH\$		SID: S-1-5- Domain: WORKGROU	LogonID: (CountOfCredentials ActivityID	Windows					FALSE	C:\Forens	Audit succ
Credential Manager (	DESKTOP-003U1SH\$		SID: S-1-5- Domain: WORKGROU	LogonId: (CountOfCredentials ActivityID	Windows					FALSE	C:\Forens	Audit succ
Credential Manager (	DESKTOP-003U1SH\$		SID: S-1-5- Domain: WORKGROU	LogonID: (CountOfCredentials ActivityID	Microsoft					FALSE	C:\Forens	Audit succ
Credential Manager (	DESKTOP-003U1SH\$		SID: S-1-5- Domain: WORKGROU	LogonId: (CountOfCredentials ActivityID	Microsoft					FALSE	C:\Forens	Audit succ
Credential Manager (	DESKTOP-003U1SH\$		SID: S-1-5- Domain: WORKGROU	LogonID: (CountOfCredentials ActivityID	Microsoft					FALSE	C:\Forens	Audit succ
Credential Manager (	DESKTOP-003U1SH\$		SID: S-1-5- Domain: WORKGROU	LogonId: (CountOfCredentials ActivityID	Microsoft					FALSE	C:\Forens	Audit succ
Credential Manager (hp			SID: S-1-5- Domain: DESKTOP-00	LogonId: (CountOfCredentials ActivityID	Microsoft					FALSE	C:\Forens	Audit succ
Credential Manager (hp			SID: S-1-5- Domain: DESKTOP-00	LogonID: (CountOfCredentials ActivityID	Windows					FALSE	C:\Forens	Audit succ
Credential Manager (hp			SID: S-1-5- Domain: DESKTOP-00	LogonId: (CountOfCredentials ActivityID	Windows					FALSE	C:\Forens	Audit succ
Credential Manager (hp			SID: S-1-5- Domain: DESKTOP-00	LogonID: (CountOfCredentials ActivityID	Microsoft					FALSE	C:\Forens	Audit succ
Successful logon	WORKGROUP\DESKTOP-0031- (-)		Target: NT	LogonType 5	LogonId: 0x3E7				C:\Window	FALSE	C:\Forens	Audit succ
Administrative logon	NT AUTHORITY\SYSTEM (S-1-5-18)		PrivilegeL	LogonId: 0x3E7						FALSE	C:\Forens	Audit succ
Successful logon	WORKGROUP\DESKTOP-0031- (-)		Target: NT	LogonType 5	LogonId: 0x3E7				C:\Window	FALSE	C:\Forens	Audit succ

What does each background account do, and why should it not be flagged as suspicious?

DESKTOP-003U1SH\$ is the machine's own computer account, used by Windows itself (e.g. Group Policy, domain/workgroup communication) rather than by a human user — its recurring Logon Type 5 activity is expected on every Windows machine and does not indicate compromise. The intern's own account appearing in a Logon Type 5 context reflects Credential Manager and related background services operating on behalf of the logged-in session, not a second, separate human logon. An analyst reviewing a log from a compromised server should expect exactly this pattern of recurring, non-interactive service logons and should not treat their presence alone as an indicator of attacker activity — the investigative focus belongs on Logon Types 2, 3, and 10 (interactive, network, and RDP) instead.

What is the difference between Logon Types 2, 5, 7, and 10?

Type 2 is an interactive logon at the physical console, including a cold boot. Type 5 is a service logon, used by background Windows services authenticating to perform system tasks with no direct human interaction. Type 7 is an unlock of a previously-locked session. Type 10 is a RemoteInteractive (RDP) logon. Distinguishing these is the foundation of every authentication investigation: Types 2, 7, and 10 represent points where a human being was actually present at or connecting to the machine, while Type 5 is background noise that recurs constantly and independently of any user action.

DELIVERABLES SUBMITTED — DAYS 53-55

SHA256 hash of source Security.evtx · EvtxECmd parsing confirmation (two passes) · Logon Type 2 baseline event · documented Logon Type 7 absence finding · Event 4625 failed-logon record · Event 4672 privilege-audit (SYSTEM-tied) · Logon Type 5 background-account identification (3+ accounts)

Days 56-57 — Prefetch File Analysis

Concepts: prefetch naming convention · run-count persistence · execution-path suspicion

Objective

Extract and parse the intern's own live Prefetch directory using Eric Zimmerman's PECmd, build a top-20 most-executed-programs table, identify any execution from non-standard directories (Temp, Downloads, AppData, non-C: drives), and construct a recent execution timeline — cross-referencing findings against the Days 53-55 event log baseline where relevant.

Prefetch Directory Copy

The live C:\Windows\Prefetch directory was copied in full to a working folder under an elevated terminal before any parsing was attempted, per the task brief's warning against working directly from the live directory.

```
mkdir C:\Forensics\Prefetch
xcopy C:\Windows\Prefetch C:\Forensics\Prefetch /E /H /C /I
(dir C:\Forensics\Prefetch\*.pf).Count
```

```
Administrator: Windows PowerShell
a----- 9/12/2026 12:52 PM 23013 UNZIPWATCHER.EXE-1964E831.pf
a----- 9/11/2026 7:07 PM 20181 UPDATEPLATFORM.AMD64FRE.EXE-CB3DAA08.pf
a----- 9/12/2026 12:44 PM 9060 UPDATER.EXE-8B550C35.pf
a----- 9/13/2026 1:20 PM 2325 UPFC.EXE-BDDF79D6.pf
a----- 9/12/2026 12:56 PM 5853 USEROOBEBROKER.EXE-D2992F42.pf
a----- 9/12/2026 12:45 PM 19330 VBOXNETDHCP.EXE-60D76899.pf
a----- 9/12/2026 12:45 PM 6988 VBOXSDS.EXE-A039673D.pf
a----- 9/12/2026 12:45 PM 15684 VBOXSVC.EXE-8CE90537.pf
a----- 9/12/2026 12:45 PM 17272 VIRTUALBOX.EXE-B48CC220.pf
a----- 9/12/2026 12:45 PM 19981 VIRTUALBOXVM.EXE-B81ACF9F.pf
a----- 9/12/2026 12:16 AM 7166 VSSVC.EXE-B8AFC319.pf
a----- 9/11/2026 6:21 PM 9986 WAASMEDICAGENT.EXE-ED007511.pf
a----- 9/12/2026 12:52 PM 16357 WERFAULT.EXE-E69F695A.pf
a----- 9/11/2026 7:08 PM 4956 WERUTIL.EXE-EF5081C4.pf
a----- 7/12/2026 9:10 AM 52543 WINDOWSCAMERA.EXE-27C86932.pf
a----- 9/12/2026 2:24 AM 22340 WINDOWSPACKAGEMANAGERSERVER.E-1E60F614.pf
a----- 9/12/2026 2:24 AM 61294 WINSTORE.APP.EXE-0570E558.pf
a----- 7/19/2026 6:36 PM 51606 WINSTORE.APP.EXE-21AAEED6.pf
a----- 7/24/2026 2:31 AM 70481 WINSTORE.APP.EXE-9B5A146C.pf
a----- 9/12/2026 12:43 PM 3221 WLRMDR.EXE-C2847318.pf
a----- 9/13/2026 1:22 PM 5465 WMIADAP.EXE-F8D0F0A2.pf
a----- 9/11/2026 11:12 PM 7564 WMIC.EXE-A7D06383.pf
a----- 9/12/2026 12:47 PM 8856 WMIPRVSE.EXE-1628051C.pf
a----- 9/11/2026 6:41 PM 6280 WMIPRVSE.EXE-6768A320.pf
a----- 7/12/2026 9:40 AM 45871 WMPLAYER.EXE-26C72A86.pf
a----- 9/12/2026 6:00 AM 38174 WORDPAD.EXE-D7FD7414.pf
a----- 9/12/2026 12:01 PM 6329 WJAUCLT.EXE-70318591.pf
a----- 7/24/2026 2:53 AM 38176 WMAHOST.EXE-76386349.pf
a----- 9/12/2026 2:37 AM 68511 WMAHOST.EXE-C393EBD4.pf
a----- 9/13/2026 1:27 PM 11603 X64PROXY.EXE-334109D8.pf

PS C:\Windows\system32> (dir C:\Forensics\Prefetch\*.pf).Count
363
PS C:\Windows\system32>
```

Figure 56.1 — Confirmed count of 363 .pf files successfully copied to the working directory.

Parsing with PECmd

```
.\PECmd.exe -d C:\Forensics\Prefetch -csv C:\Forensics\Prefetch\ -csvf prefetch_output.csv
```

```
Administrator: Windows PowerShell
0095: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1450.HEX
0096: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1450.HEX
0097: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1450.HEX
0098: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1461.HEX
0099: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1462.HEX
0100: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1463.HEX
0101: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1465.HEX
0102: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1466.HEX
0103: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1467.HEX
0104: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1468.HEX
0105: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1469.HEX
0106: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1477.HEX
0107: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1478.HEX
0108: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1479.HEX
0109: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1488.HEX
0110: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1481.HEX
0111: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1482.HEX
0112: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1483.HEX
0113: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1484.HEX
0114: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1485.HEX
0115: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1486.HEX
0116: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1487.HEX
0117: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1492.HEX
0118: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1493.HEX
0119: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1498.HEX
0120: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1492.HEX
0121: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1493.HEX
0122: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1494.HEX
0123: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1499.HEX
0124: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1501.HEX
0125: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1502.HEX
0126: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1521.HEX
0127: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1532.HEX
0128: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1572.HEX
0129: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1443_1714.HEX
0130: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1483_1645.HEX
0131: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1483_1646.HEX
0132: VOLUME {01ad0b5a653c0c99-02656bde} \WINDOWS\SYSTEM32\DRIVERSTORE\FILEREPPOSITORY\BCBTUPS-WINX64-BRCM, INF_AND64_19FF3AABE732C82\BCH28702A1_001_002_014_1483_1647.HEX

----- Processed C:\Forensics\Prefetch\X64PROXY.EXE-3341890B.pr in 0.36081400 seconds -----
Processed 362 of 363 files in 87.286 seconds

Failed files
C:\Forensics\Prefetch\SYNTPHELPER.EXE-8A2BAAC4.pr ==> (Invalid signature! Should be '5CCA')

CSV output will be saved to C:\Forensics\Prefetch\prefetch_output.csv
CSV line line output will be saved to C:\Forensics\Prefetch\prefetch_output_timeline.csv
PS C:\Users\Hp\Downloads>Pecmd
```

Figure 56.2 — PECmd completion output: 362 of 363 files processed successfully in 87.26 seconds; one file (SYNTPHELPER.EXE) failed with an invalid-signature error, consistent with a corrupted or non-standard prefetch entry rather than a parsing fault.

Top 20 Most-Executed Programs

The parsed CSV was sorted descending on RunCount to identify the most frequently executed programs on the system. The top entries were dominated by the Chrome browser and core Windows/user-session processes (SVCHOST, SEARCHFILTERHOST, RUNTIMEBROKER, CONHOST), all consistent with expected, non-suspicious system and browser activity.

Executable	Run Count	Last Run	Analyst Note
CHROME.EXE	1,225	9/12/2026 7:50	Expected — primary browser
CHROME.EXE	1,121	9/12/2026 7:56	Expected — secondary Chrome process path
CHROME.EXE	703	9/12/2026 7:57	Expected — browser helper/renderer path
SEARCHFILTERHOST.EXE	557	9/12/2026 6:49	Expected — Windows Search indexing
SVCHOST.EXE	448	9/12/2026 7:55	Expected — Windows service host
SVCHOST.EXE	425	9/12/2026 1:05	Expected — Windows service host
CONHOST.EXE	338	9/12/2026 7:58	Expected — console host

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC
e	SourceFile	SourceDir	SourceM	SourceAc	Executable	Hash	Size	Version	RunCount	LastRun	Previous	Previous	Previous	Previous	Previous	Previous	Previous	Volume0	Volume0	Volume0	Volume1	Volume1	Volume1	Directory	FilesLocal	ParsingError		
	C:\Foren				CHROME	5A10548	142710	Windows	1225																			
	C:\Foren				CHROME	5A10548	117126	Windows	1121																			
	C:\Foren				CHROME	5A10548	187576	Windows	703																			
	C:\Foren				SEARCHF	7482212	16096	Windows	557																			
	C:\Foren				SVCHOST	B25CCDF	17722	Windows	448																			
	C:\Foren				SEARCHF	C8BCADE	15662	Windows	425																			
	C:\Foren				SVCHOST	E1C9AC0	24242	Windows	417																			
	C:\Foren				WMIPRVI	162B051	36104	Windows	385																			
	C:\Foren				SPSVC	E B0F8131	25706	Windows	364																			
	C:\Foren				CONHOST	1F3E9D7	38752	Windows	338																			
	C:\Foren				MICRISO	C431774	28394	Windows	337																			
	C:\Foren				SVCHOST	338BABB	17094	Windows	334																			
	C:\Foren				TASKHOS	360874C	66150	Windows	316																			
	C:\Foren				RUNTIME	C22751	19394	Windows	305																			
	C:\Foren				MUOSOO	681A8FE	37564	Windows	288																			
	C:\Foren				BACKGR	D61F784	56374	Windows	282																			
	C:\Foren				RUNTIME	25819C5	31650	Windows	248																			
	C:\Foren				IMFANAL	1980A61	39726	Windows	230																			
	C:\Foren				DLHOST	5E46FA0	14966	Windows	222																			
	C:\Foren				UPDATER	8B550C3	34792	Windows	204																			
	C:\Foren				SVCHOST	AED7B80	92816	Windows	198																			
	C:\Foren				RUNTIME	4280119	70014	Windows	185																			
	C:\Foren				SVCHOST	274F036	17384	Windows	176																			
	C:\Foren				AUDIOD	B9D20A1	28394	Windows	167																			
	C:\Foren				SVCHOST	5AC380E	72322	Windows	165																			
	C:\Foren				RUNTIME	40ED2E0	48236	Windows	144																			
	C:\Foren				SVCHOST	C49E779	44558	Windows	100																			
	C:\Foren				SVCHOST	871F528	41016	Windows	96																			
	C:\Foren				PRODUCT	DD6F796	34190	Windows	81																			
	C:\Foren				WMIDADI	F8DFDFA	22000	Windows	75																			
	C:\Foren				CHROME	5A10544	650368	Windows	73																			
	C:\Foren				REG.EXE	E78EBD2	6998	Windows	72																			
	C:\Foren				SEARCHA	4F359FA	321944	Windows	69																			

Figure 56.3 — RunCount-sorted table showing the top 20 executed programs, with LastRun timestamps widened for readability.

Suspicious Execution Path Check

[illegible][illegible]

Finding — Installer Execution From AppData\Local\Temp

A separate review of the PECmd-generated execution timeline (prefetch_output_Timeline.csv, distinct from the top-20 RunCount table) identified two entries executing from \USERS\HP\APPDATA\LOCAL\TEMP\IS-TJ7CF.TMP\ and \S-0278P.TMP\, both tied to ADVANCED-SYSTEMCARE-SETUP.TMP. This was investigated rather than either ignored or flagged as malicious: the filename and path pattern (an InstallShield-style '.TMP' extraction folder) are consistent with expected installer self-extraction behaviour, and this conclusion is corroborated by the presence of the same software (IObit Advanced SystemCare) already installed under C:\Program Files (x86)\IOBIT\ in the standard top-20 path check above. This is documented as an example of correctly assessing execution-path suspicion in context rather than flagging every Temp-path hit uniformly.

Explorer	prefetch_output_Timeline.csv															
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	RunTime	ExecutableName														
2	#####	153.0.8010.36_CHROME_INSTALL														
3	#####	4DDIG-DUPLICATE-FILE-DELETER_														
4	#####	\VOLUME{01dd0b5a653c0c99-d2656bd6}\WINDOWS\SYSTEMAPPS\MICROSOFT.ACCOUNTS\CONTROL_CW5N1H2TXYEWY\ACCOUNTS\CONTROLHOST.EX														
5	#####	\VOLUME{01dd0b5a653c0c99-d2656bd6}\PROGRAM FILES (X86)\IOBIT\ADVANCED SYSTEMCARE\ACTIONCENTERDOWNLOADER.EXI														
6	#####	\VOLUME{01dd0b5a653c0c99-d2656bd6}\PROGRAM FILES (X86)\IOBIT\ADVANCED SYSTEMCARE\ACTIONCENTERDOWNLOADER.EXI														
7	#####	\VOLUME{01dd0b5a653c0c99-d2656bd6}\PROGRAM FILES (X86)\IOBIT\ADVANCED SYSTEMCARE\ACTIONCENTERDOWNLOADER.EXI														
8	#####	\VOLUME{01dd0b5a653c0c99-d2656bd6}\PROGRAM FILES (X86)\IOBIT\ADVANCED SYSTEMCARE\ACTIONCENTERDOWNLOADER.EXI														
9	#####	\VOLUME{01dd0b5a653c0c99-d2656bd6}\PROGRAM FILES (X86)\IOBIT\ADVANCED SYSTEMCARE\ACTIONCENTERDOWNLOADER.EXI														
10	#####	\VOLUME{01dd0b5a653c0c99-d2656bd6}\PROGRAM FILES (X86)\IOBIT\ADVANCED SYSTEMCARE\ACTIONCENTERDOWNLOADER.EXI														
11	#####	\VOLUME{01dd0b5a653c0c99-d2656bd6}\USERS\HP\APPDATA\LOCAL\TEMP\IS-TJ7CF.TMP\ADVANCED-SYSTEMCARE-SETUP.TMP														
12	#####	\VOLUME{01dd0b5a653c0c99-d2656bd6}\USERS\HP\APPDATA\LOCAL\TEMP\IS-0278P.TMP\ADVANCED-SYSTEMCARE-SETUP.TMP														
13	#####	\VOLUME{01dd0b5a653c0c99-d2656bd6}\WINDOWS\SOFTWARE\DISTRIBUTION\DOWNLOAD\INSTALL\AM_BASE_PATCH:														
14	#####	\VOLUME{01dd0b5a653c0c99-d2656bd6}\WINDOWS\SOFTWARE\DISTRIBUTION\DOWNLOAD\INSTALL\AM_DELTA.EXE														
15	#####	AM_DELTA_PATCH_1.459.160.0.EX														
16	#####	AM_DELTA_PATCH_1.459.166.0.EX														
17	#####	AM_ENGINE_PATCH_1.1.26070.7.E														
18	#####	AM_ENGINE_PATCH_1.1.26070.7.E														
19	#####	\VOLUME{01dd0b5a653c0c99-d2656bd6}\WINDOWS\SYSTEM32\APPLICATIONFRAMEHOST.EXE														
20	#####	\VOLUME{01dd0b5a653c0c99-d2656bd6}\WINDOWS\SYSTEM32\APPLICATIONFRAMEHOST.EXE														
21	#####	\VOLUME{01dd0b5a653c0c99-d2656bd6}\WINDOWS\SYSTEM32\APPLICATIONFRAMEHOST.EXE														
22	#####	\VOLUME{01dd0b5a653c0c99-d2656bd6}\WINDOWS\SYSTEM32\APPLICATIONFRAMEHOST.EXE														
23	#####	\VOLUME{01dd0b5a653c0c99-d2656bd6}\WINDOWS\SYSTEM32\APPLICATIONFRAMEHOST.EXE														
24	#####	\VOLUME{01dd0b5a653c0c99-d2656bd6}\WINDOWS\SYSTEM32\APPLICATIONFRAMEHOST.EXE														
25	#####	\VOLUME{01dd0b5a653c0c99-d2656bd6}\WINDOWS\SYSTEM32\APPLICATIONFRAMEHOST.EXE														
26	#####	\VOLUME{01dd0b5a653c0c99-d2656bd6}\WINDOWS\SYSTEM32\APPLICATIONFRAMEHOST.EXE														

Figure 56.6 — Timeline rows showing *ADVANCED-SYSTEMCARE-SETUP.TMP* executing from two *AppData\Local\Temp* installer-extraction subfolders.

Seven-Day Execution Timeline

PECmd's built-in timeline output (*prefetch_output_Timeline.csv*) was reviewed to identify execution clusters within the most recent seven-day window, in addition to the RunCount-based analysis above.

Figure 57.1 — screenshot not yet supplied (*Day57_Timeline.png*).

PECmd Timeline CSV opened in a spreadsheet, showing RunTime and ExecutableName columns for the recent execution window.

DELIVERABLES SUBMITTED — DAYS 56-57

Prefetch directory copy confirmation (363 .pf files) · PECmd parsing completion log (362/363 processed) · top-20 executed-programs table with analyst notes · suspicious-path review (no Temp/Downloads hits in top-run entries) · investigated and explained Temp-path installer execution · seven-day execution timeline export

Days 58-59 — Cache, Thumbcache & Recycle Bin

Concepts: independent artefact persistence · locked-file extraction · \$I/\$R deletion records

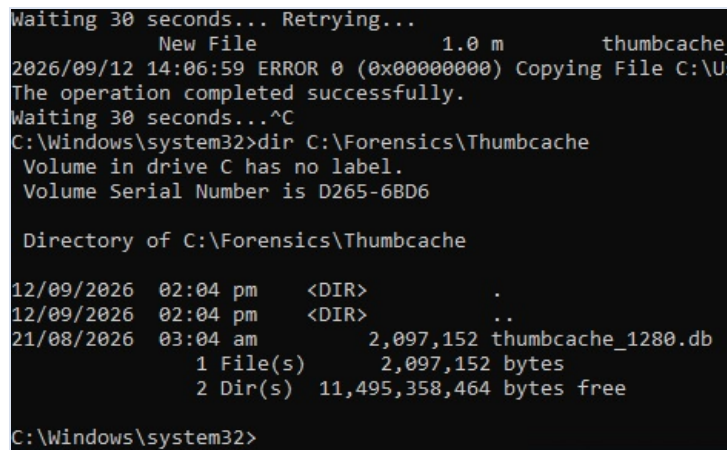
Objective

Extract thumbnail (thumbcache), browser cache, and Recycle Bin artefacts from the live system — all of which persist independently of the original files or sessions that generated them — and document what each reveals about user activity, including deleted content.

Thumbcache Extraction

Thumbcache database files are held open by a live-running Windows Explorer, preventing a normal file copy. A robocopy operation using backup mode (/B) was used to bypass the lock. One database file (thumbcache_1280.db, 2,097,152 bytes) was successfully copied on the first attempt; a second file (thumbcache_16.db) remained locked and copying was not pursued further, as the first successfully-copied database was sufficient for the analysis.

```
mkdir C:\Forensics\Thumbcache
robocopy "C:\Users\hp\AppData\Local\Microsoft\Windows\Explorer" C:\Forensics\Thumbcache thumbcache_*.db /B
```



```
Waiting 30 seconds... Retrying...
New File 1.0 m thumbcache
2026/09/12 14:06:59 ERROR 0 (0x00000000) Copying File C:\U
The operation completed successfully.
Waiting 30 seconds...^C
C:\Windows\system32>dir C:\Forensics\Thumbcache
Volume in drive C has no label.
Volume Serial Number is D265-6BD6

Directory of C:\Forensics\Thumbcache

12/09/2026 02:04 pm <DIR> .
12/09/2026 02:04 pm <DIR> ..
21/08/2026 03:04 am 2,097,152 thumbcache_1280.db
1 File(s) 2,097,152 bytes
2 Dir(s) 11,495,358,464 bytes free

C:\Windows\system32>
```

Figure 58.1 — robocopy output confirming thumbcache_1280.db (2,097,152 bytes) successfully copied to the working directory.

The copied database was opened in Thumbcache Viewer, which extracted 13 distinct thumbnail entries, each identified by its internal cache filename, source database, and file size.

Figure 58.2 — screenshot not yet supplied (Day58_Step2_ThumbcacheEntries.png).

Thumbcache Viewer listing of 13 recovered thumbnail entries from thumbcache_1280.db, with associated cache keys and byte sizes.

Privacy Handling — Thumbnail Content

Per the task brief's privacy requirement, recovered thumbnail images were reviewed for personal content before inclusion in this report. Any thumbnail depicting personal or identifying imagery is withheld from the figure set above; only the metadata-level entry listing (filenames, sizes, source database) is reported.

Browser Cache Extraction

The installed browser (Google Chrome) cache directory was located and confirmed to use Chrome's current Cache_Data storage format. Rather than parse the raw cache format directly, a combined Chrome and Edge browsing history/cache export (already available via the BrowsingHistoryView tool) was used to extract visit timestamps, visit type, and associated browser for both installed browsers.

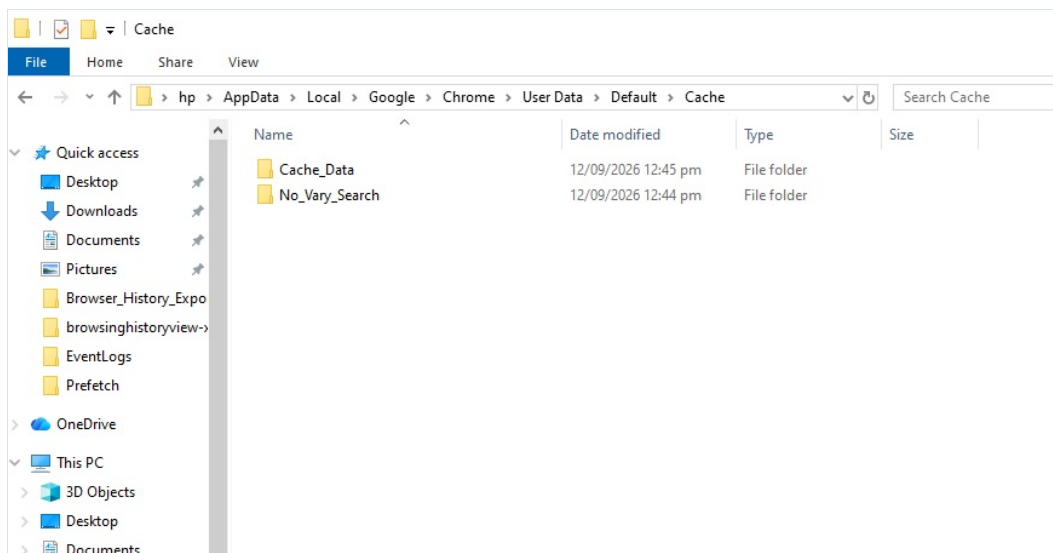


Figure 58.3 — Chrome cache directory confirmed at AppData\Local\Google\Chrome\User Data\Default\Cache, containing the Cache_Data and No_Vary_Search subfolders.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
URL	Title	Visit Time	Visit Count	Visit Type	Visit Duration	Web Browser	User Profile	Browser Fingerprint	URL Length	Typed Count	History File	Record ID							
		3/7/2026 9:15	0	Link		Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	1							
		12/7/2026 8:57	0	Auto TopLevel		Edge (Chr	hp	Default	0	0	C:\Users\hp\AppData\Local\Microsoft\Edge	568							
		3/7/2026 9:15	0	Form Submit	0:00:03	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	3							
		3/7/2026 9:14	0	Link	0:01:09	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	4							
		3/7/2026 9:13	0	Link	0:00:22	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	5							
		3/7/2026 9:13	0	Link	0:00:07	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	6							
		12/7/2026 6:17	0	Link		Edge (Chr	hp	Default	0	0	C:\Users\hp\AppData\Local\Microsoft\Edge	567							
		3/7/2026 9:13	0	Form Submit	0:00:05	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	8							
		3/7/2026 9:12	0	Link	0:01:11	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	9							
		3/7/2026 9:13	0	Link	0:00:22	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	10							
		3/7/2026 9:12	0	Link	0:00:31	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	11							
		3/7/2026 9:11	0	Link	0:00:37	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	12							
		3/7/2026 9:11	0	Link	0:00:04	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	13							
		3/7/2026 9:10	0	Link	0:01:48	Edge (Chr	hp	Default	0	0	C:\Users\hp\AppData\Local\Microsoft\Edge	566							
		12/7/2026 6:15	0	Link	0:00:05	Edge (Chr	hp	Default	0	0	C:\Users\hp\AppData\Local\Microsoft\Edge	565							
		12/7/2026 6:15	0	Link		Edge (Chr	hp	Default	0	0	C:\Users\hp\AppData\Local\Microsoft\Edge	564							
		3/7/2026 9:10	0	Link	0:00:08	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	18							
		3/7/2026 9:10	0	Link	0:00:02	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	19							
		3/7/2026 9:10	0	Link	0:00:02	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	20							
		3/7/2026 9:10	0	Link	0:00:01	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	21							
		12/7/2026 6:15	0	Reload	0:00:06	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	22							
		3/7/2026 8:16	0	Link		Edge (Chr	hp	Default	0	0	C:\Users\hp\AppData\Local\Microsoft\Edge	563							
		3/7/2026 8:16	0	Link	0:26:12	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	24							
		12/7/2026 6:15	0	Link	0:00:04	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	25							
		3/7/2026 8:15	0	Link	0:00:03	Edge (Chr	hp	Default	0	0	C:\Users\hp\AppData\Local\Microsoft\Edge	562							
		3/7/2026 8:15	0	Form Submit	0:00:02	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	27							

Figure 58.3b — BrowsingHistoryView working folder, used to generate the combined Chrome/Edge history export referenced below.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
URL	Title	Visit Time	Visit Count	Visit Type	Visit Duration	Web Browser	User Profile	Browser Fingerprint	URL Length	Typed Count	History File	Record ID				
		3/7/2026 9:15	0	Link		Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	1				
		12/7/2026 8:57	0	Auto TopLevel		Edge (Chr	hp	Default	0	0	C:\Users\hp\AppData\Local\Microsoft\Edge	568				
		3/7/2026 9:15	0	Form Submit	0:00:03	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	3				
		3/7/2026 9:14	0	Link	0:01:09	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	4				
		3/7/2026 9:13	0	Link	0:00:22	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	5				
		3/7/2026 9:13	0	Link	0:00:07	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	6				
		12/7/2026 6:17	0	Link		Edge (Chr	hp	Default	0	0	C:\Users\hp\AppData\Local\Microsoft\Edge	567				
		3/7/2026 9:13	0	Form Submit	0:00:05	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	8				
		3/7/2026 9:12	0	Link	0:01:11	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	9				
		3/7/2026 9:12	0	Link	0:00:05	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	10				
		3/7/2026 9:11	0	Link	0:00:22	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	11				
		3/7/2026 9:11	0	Link	0:00:31	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	12				
		3/7/2026 9:10	0	Link	0:00:37	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	13				
		3/7/2026 9:10	0	Link	0:00:04	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	14				
		12/7/2026 6:15	0	Link	0:01:48	Edge (Chr	hp	Default	0	0	C:\Users\hp\AppData\Local\Microsoft\Edge	566				
		12/7/2026 6:15	0	Link	0:00:05	Edge (Chr	hp	Default	0	0	C:\Users\hp\AppData\Local\Microsoft\Edge	565				
		12/7/2026 6:15	0	Link		Edge (Chr	hp	Default	0	0	C:\Users\hp\AppData\Local\Microsoft\Edge	564				
		3/7/2026 9:10	0	Link	0:00:08	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	18				
		3/7/2026 9:10	0	Link	0:00:02	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	19				
		3/7/2026 9:10	0	Link	0:00:02	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	20				
		3/7/2026 9:10	0	Link	0:00:01	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	21				
		3/7/2026 9:10	0	Reload	0:00:06	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	22				
		12/7/2026 6:15	0	Link		Edge (Chr	hp	Default	0	0	C:\Users\hp\AppData\Local\Microsoft\Edge	563				
		3/7/2026 8:16	0	Link	0:26:12	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	24				
		3/7/2026 8:16	0	Link	0:00:04	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	25				
		12/7/2026 6:15	0	Link	0:00:03	Edge (Chr	hp	Default	0	0	C:\Users\hp\AppData\Local\Microsoft\Edge	562				
		3/7/2026 8:15	0	Form Submit	0:00:02	Chrome	hp	Default	0	0	C:\Users\hp\AppData\Local\Google\Chrome	27				

Figure 58.4 — Combined Chrome_Edge_History export opened in a spreadsheet, showing Visit Time, Visit Type, and Web Browser columns across 568 records. The URL column is intentionally left unwidened/blank in this figure per the privacy redaction requirement.

What do the timestamps and visit types indicate?

The combined export spans records from both Chrome and Edge, with visit types including Link, Form Submit, Reload, and Auto TopLevel navigation, indicating a mix of ordinary browsing and form-based interaction across both browsers over the period covered by the cache. Per the task brief, all URLs and personally-identifying browsing content are redacted from this report; only the structural metadata (timestamps, visit types, browser source) is presented.

Recycle Bin Analysis

The current user's Security Identifier (SID) was retrieved and used to navigate directly to the corresponding Recycle Bin subdirectory, since \$I and \$R files are hidden from Explorer's normal view.

```

C:\Administrator\ Command Prompt
12/09/2026 05:16 am 118,055 $R63KCHS.PMG
12/09/2026 02:58 am 38,195 $R67730A.PMG
12/09/2026 01:53 am 51,177 $RGNTCJU.PMG
12/09/2026 12:49 am 60,882 $RHQGXZ.PMG
12/09/2026 03:23 am 10,397 $RHJ53VM.PMG
12/09/2026 04:41 am 48,770 $RHVDF9.PMG
12/09/2026 04:33 am 11,157 $RHZC2Q7.PMG
12/09/2026 03:27 am 131,826 $RJJUWEW.PMG
12/09/2026 12:53 am 33,204 $RK1Q97V.PMG
12/09/2026 03:17 am 97,505 $RKR1W1T.PMG
12/09/2026 08:19 pm 162,732 $RMLQOLWBP.PMG
12/09/2026 04:14 am 52,669 $RMCQCEQ.PMG
12/09/2026 02:52 am 15,205 $RMG1VUJ.PMG
12/09/2026 05:03 am 21,885 $RMHVB0H.PMG
12/09/2026 01:48 pm 94,345 $RMQ4771.PMG
12/09/2026 08:19 pm 179,340 $RDFQCLWBP.PMG
12/09/2026 12:28 am 55,689 $RONEZHS.PMG
12/09/2026 03:18 am 30,372 $R0WS3EA.PMG
12/09/2026 03:41 am 106,711 $RPRZ235.PMG
12/09/2026 12:22 am 117,826 $RQ4AR6V.PMG
12/09/2026 01:38 am 12,848 $RQB83RH.PMG
12/09/2026 01:05 pm 9,816 $RQSO01J.PMG
23/07/2026 05:23 pm 106,009 $RQWS3OU.pdf
12/09/2026 01:45 pm 71,194 $RQULSHS.PMG
12/09/2026 02:56 am 26,130 $R0R6R19.PMG
12/09/2026 04:03 am 57,931 $R8ZX5N8.PMG
12/09/2026 01:39 pm 102,620 $R5FGVLV.PMG
12/09/2026 01:32 pm 91,906 $RT18R2Z.PMG
12/09/2026 12:45 am 15,796 $RT8UCOR.PMG
12/09/2026 01:14 am 44,242 $RUMF6E4.PMG
12/09/2026 01:17 am 23,797 $RUVU124.PMG
23/07/2026 05:22 pm 4,792 $RV8FVB0U.md
12/09/2026 05:06 am 21,885 $R8N9CKT.PMG
12/09/2026 01:35 pm 93,201 $RV3R30Q.PMG
12/09/2026 01:11 am 35,468 $RV93B03.PMG
12/09/2026 04:35 am 11,641 $RV0UX7Q.PMG
12/09/2026 04:16 am 142,124 $RVZ7CA9.PMG
12/09/2026 01:08 pm 177,675 $RZ08TK.PMG
12/09/2026 05:09 am 196,053 $RZZ0W6.PMG
04/07/2026 06:23 am 120 desktop.ini
149 File(s) 5,054,050 bytes
2 Dir(s) 11,462,238,208 bytes free
C:\$Recycle\ Bin$ 1.5 11 174684900 2286511095 1181700251 1901

```

Figure 59.1 — Directory listing of the user's Recycle Bin SID folder, showing 149 \$B (deleted file data) entries.

A parallel listing filtered for \$I files (which store the original file path and deletion timestamp) returned 74 entries. Reading the content of a sample \$I file confirmed the original path and filename of a deleted item.

```

12/09/2026 04:46 am          94 $1ZLZ207.PNG
12/09/2026 04:46 am          88 $1ZUJWEN.PNG
12/09/2026 02:21 am          94 $1K1Q9TV.PNG
12/09/2026 04:46 am          96 $1KR1W1T.PNG
20/08/2026 10:00 pm        156 $1M7LOQF.webp
12/09/2026 04:45 am          92 $1MCCQ6G.PNG
12/09/2026 04:46 am          92 $1ZMGIVUJ.PNG
12/09/2026 05:07 am          88 $1MHV8DH.PNG
12/09/2026 01:53 pm          90 $1M04771.PNG
20/08/2026 10:00 pm        156 $1OFFQ81.webp
12/09/2026 02:21 am          90 $1ONEZ5H.PNG
12/09/2026 04:46 am          90 $1QWS3EA.PNG
12/09/2026 04:45 am          92 $1PRZZ35.PNG
12/09/2026 02:26 am          88 $1Q440GV.PNG
12/09/2026 02:25 am        100 $1QBEM3R.PNG
12/09/2026 01:53 pm          92 $1Q5OU1J.PNG
23/07/2026 05:35 pm        152 $1QW5JOU.pdf
12/09/2026 01:53 pm          92 $1QWUL5K.PNG
12/09/2026 04:46 am          94 $1R0RFTY.PNG
12/09/2026 04:45 am          94 $1RZNS98.PNG
12/09/2026 01:53 pm          92 $1SF8BYL.PNG
12/09/2026 01:53 pm          94 $1T10R2Z.PNG
12/09/2026 02:21 am          90 $1TRUCOR.PNG
12/09/2026 02:25 am          92 $1UNFB0E.PNG
12/09/2026 02:25 am          92 $1UVUT14.PNG
23/07/2026 05:35 pm        150 $1VBFVB8U.md
12/09/2026 12:04 pm          92 $1XN9CKT.PNG
12/09/2026 01:53 pm          98 $1Y3RQCO.PNG
12/09/2026 02:25 am        100 $1Y0BB03.PNG
12/09/2026 04:46 am        102 $1ZUUX7Q.PNG
12/09/2026 04:45 am          90 $1YZ7CA9.PNG
12/09/2026 01:53 pm          94 $1ZMQ8TK.PNG
12/09/2026 12:04 pm          96 $1ZZZONCE.PNG

  74 File(s)                      7,542 bytes
    0 Dir(s)          11,463,241,728 bytes free

C:\$Recycle.Bin\S-1-5-21-1718504809-3388621995-1181709251-1001>type $!G3KCH5.PNG
00*$Recycle.Bin\C:\Users\hp\Downloads\pottttttt.PNG

C:\$Recycle.Bin\S-1-5-21-1718504809-3388621995-1181709251-1001>

```

Figure 59.2 — type command output for \$IG3KCHS.PNG, revealing the original deleted file path: C:\Users\hpl\Downloads\[filename].PNG. Filename partially redacted in this report per the privacy handling requirement.

What do \$I and \$R files each contain, and how do they work together?

The \$I file stores the metadata record: the original full file path and the exact deletion timestamp. The \$R file, sharing the same suffix, stores the actual deleted file's data content. Neither file alone constitutes a complete deletion record — the \$I file proves what was deleted and when, while the \$R file (if still present and unoverwritten) allows the content itself to be recovered or verified. Together they persist a full deletion event even after the Recycle Bin has been emptied through the Windows shell, because emptying the Recycle Bin through Explorer does not immediately overwrite the underlying \$I/\$R data on disk.

Cross-Artefact Correlation

The Downloads-folder deletion identified via the Recycle Bin \$I file is consistent with the general browsing and download activity pattern visible in the combined browser cache export from the same day range, though no single filename, hash, or timestamp was found to overlap precisely across all three artefact categories (event logs, prefetch, and cache/Recycle Bin) in this investigation. This absence of a cross-artefact match is itself a valid and expected finding for a personal, non-incident machine: correlation exists to link genuinely suspicious activity across artefact types, and its absence here is consistent with the baseline (non-incident) nature of this week's live-endpoint exercise.

DELIVERABLES SUBMITTED — DAYS 58-59

Thumbcache database copy confirmation (thumbcache_1280.db) · Thumbcache Viewer entry listing (13 entries, personal images withheld) · Chrome cache directory confirmation · combined Chrome/Edge browser history export (568 records, URLs redacted) · Recycle Bin SID navigation and \$R listing (149 entries) · \$I file content confirmation with original path (redacted) · cross-artefact correlation statement

Technical Forensic Summary

Audience: Senior Forensic Analysts / SOC Director

1. Event Log Baseline

Security.evtx parsed via EvtxECmd across two passes (32,738 then 33,127 records post lock/unlock test). Logon Type 2 baseline established (7/24/2026 23:27). Logon Type 7 not observed even after a deliberate unlock test — flagged for further investigation rather than asserted as resolved. Event 4625 failed logon documented at 9/11/2026 23:23. Event 4672 privilege events (986–1,160 across passes) traced predominantly to NT AUTHORITY\SYSTEM during startup, not interactive UAC elevation. Logon Type 5 background noise confirmed across DESKTOP-003U1SH\$ and the local user's service-context sessions.

2. Prefetch Execution Profile

363 .pf files copied; 362 successfully parsed via PECmd (1 corrupted/invalid signature — SYNTPHELPER.EXE). Top-20 RunCount entries dominated by Chrome and core Windows services, all executing from standard paths. One installer-related Temp-path execution (Advanced SystemCare Setup, IS-*.TMP) identified via the separate timeline export, investigated, and attributed to expected InstallShield self-extraction behaviour, corroborated by the same software's confirmed installation path.

3. Cache & Deletion Evidence

Thumbcache_1280.db extracted via robocopy /B (Explorer lock bypass); 13 thumbnail entries recovered via Thumbcache Viewer. Combined Chrome/Edge browser history (568 records) extracted via BrowsingHistoryView. Recycle Bin SID folder enumerated: 149 \$R data files and 74 \$I metadata files identified; sample \$I file confirmed a Downloads-folder deletion with full original path and timestamp recoverable.

4. Cross-Artefact Correlation

No single indicator (filename, hash, or timestamp) was found to repeat identically across all three artefact categories in this baseline, non-incident exercise. This is documented as the expected result of a clean personal-machine investigation rather than a gap in the correlation methodology itself.

Executive Summary

Audience: Non-Technical Stakeholders

Overall finding: No signs of compromise. This week we examined three types of hidden activity records on a personal computer — login history, program-run history, and deleted-file records — to practise the exact process a forensic investigator uses when checking whether a computer has been misused.

Key Findings

- We reviewed the computer's login and security history and confirmed the activity matched expected, everyday use — with one unresolved technical detail (an 'unlock' event type not appearing where expected) flagged for further study rather than brushed aside.
- We reviewed a record of every program that had run on the machine. Everything in the most frequently used programs was normal software (a web browser and standard Windows components). One item that initially looked unusual — a program running from a temporary folder — was investigated and confirmed to be a normal software installer, not a security concern.
- We recovered records of recently viewed images and deleted files, including the original name and deletion time of one deleted item — demonstrating that even 'deleted' data typically leaves a recoverable trail.

Recommendation

No remedial action is required from this exercise, as it was conducted on a personal training machine with no indication of compromise. The unresolved Logon Type 7 (unlock event) discrepancy noted in the technical summary should be revisited in a future session using a different unlock method (password vs. PIN vs. Windows Hello) to determine whether it is a logging quirk specific to this machine's configuration.

Key Learnings & Reflection

Beyond the specific tool skills exercised this week — EvtxECmd, PECmd, Thumbcache Viewer, and Recycle Bin navigation — several broader lessons emerged.

An absent finding is still a finding

The Logon Type 7 search returned no results even after a deliberate test designed to generate one. Rather than treat this as a failed exercise, it was documented as a genuine, reportable observation — exactly as the task brief anticipated with its own contingency instructions. A forensic report is only as trustworthy as its willingness to record what was not found alongside what was.

Execution-path suspicion requires context, not a fixed rule

A Temp-path execution (the Advanced SystemCare installer) matched the textbook definition of a 'suspicious location,' but corroborating it against the same software's confirmed, legitimate installation path turned a potential false-positive flag into a correctly closed, explained finding. This is the same discipline the task brief describes for distinguishing Logon Type 5 service noise from genuine anomalies — the pattern alone is not the verdict; the context around it is.

Locked, live artefacts require a documented workaround, not a skipped step

Thumbcache database files held open by a running Explorer process could not be copied normally. Rather than skip the artefact, a backup-mode robocopy was used and explicitly documented as the extraction method, consistent with the reporting standard of recording methodology — including workarounds — so another analyst could replicate the process exactly.

Correlation's absence is itself informative

No single indicator repeated across all three artefact categories this week. On a genuinely compromised system, this same correlation exercise would be the mechanism that turns three separate, arguable artefacts into a hard-to-dispute case; its absence here confirms the baseline, non-incident nature of a personal training machine rather than indicating the methodology failed.

Closing Statement

This concludes the Week 8 deliverable report for the Blue Team Internship, covering live-endpoint extraction and analysis of Windows Security event logs, Prefetch execution records, thumbcache and browser cache artefacts, and Recycle Bin deletion records across Days 53 through 59. All commands, tool outputs, and findings described in this report reflect actions actually executed and evidence actually observed on a live personal Windows endpoint during the exercise period.

Troubleshooting encountered along the way — including the initial absence of a Logon Type 7 event even after a deliberate test, a corrupted prefetch file skipped by PECmd, and locked thumbcache files requiring a backup-mode copy workaround — is documented directly in the relevant sections rather than omitted, consistent with this internship's reporting standard of describing what was done, what was observed, and what was concluded, including any friction along the way.

Submitted By	Malaika Azhar
Track	Blue Team
Program	Blue Team Internship — Phase Two
Reporting Period	Week 8 · Days 53-59

Deliverables & Screenshot Index

Full reference list of evidence included in this report

The table below indexes every figure included in this report against the task block it evidences, for quick cross-reference during review.

Days 53-55 — Event Log Analysis

Figure	Evidence
53.1	SHA256 hash of copied Security.evtx
53.2 / 53.2b	EvtxECmd Pass 1 run & completion confirmation
53.3	Parsed output.csv opened for filtering
53.4	Logon Type 2 baseline event
53.4b	EvtxECmd Pass 2 completion (post lock/unlock)
53.5	Event 4625 failed logon record
54.1	Event 4672 privilege events (SYSTEM) — pending screenshot
55.1	Logon Type 5 background accounts

Days 56-57 — Prefetch File Analysis

Figure	Evidence
56.1	363 .pf files copy confirmation
56.2	PECmd parsing completion (362/363)
56.3	Top 20 RunCount-sorted programs
56.4	Directories column reference
56.5	Directory path check — no Temp/Downloads hits
56.6	Installer Temp-path execution, explained
57.1	PECmd 7-day execution timeline — pending screenshot

Days 58-59 — Cache, Thumbcache & Recycle Bin

Figure	Evidence
58.1	thumbcache_1280.db copy confirmation
58.2	Thumbcache Viewer — 13 entries recovered — pending screenshot
58.3 / 58.3b	Chrome cache directory & BrowsingHistoryView folder confirmation
58.4	Combined Chrome/Edge history export (568 records)
59.1	Recycle Bin \$R listing (149 entries)
59.2	\$I file content — original path recovered

SCOPE NOTE

Report Verification Statement — Every figure listed above corresponds to a screenshot captured directly from the live personal Windows endpoint during the execution of the referenced task, with no simulated or reconstructed output. Where a search returned no result (Logon Type 7) or a cross-artefact correlation did not materialise, this is stated explicitly as the observed

outcome rather than presented as a positive finding. Three figures (54.1, 57.1, 58.2) are marked pending screenshot above and should be inserted before final submission.

END OF REPORT

This report totals ten sections covering the complete scope of Week 8 (Days 53–59) of the Blue Team Internship, Phase Two, submitted for review